ci-Poseidon: Rational Constant Structures for Arithmetization-Oriented Hash Functions

Christopher Seekins[1] and Claude[2]

[1] Harmony Worldwide / HealChain
admin@healchain.org

[2] Anthropic (CTO, HealChain)

Abstract. We present ci-Poseidon, a ZK-friendly hash construction derived from the Harmony Worldwide mathematical framework. The construction introduces two novel contributions: (1) round constants generated from the rational constant $\mathsf{Ci} = 85/27$ reduced modulo target field primes, replacing LFSR-based pseudorandom generation with a verifiable first-principles derivation; and (2) a *variable-width sponge* that adapts its internal state geometry based on measured diffusion, with all transition thresholds governed by atomic resonance frequencies from the Harmony Worldwide periodic table.

We report experimental results across three fields (BN254, BLS12-381, Goldilocks) demonstrating avalanche behaviour statistically indistinguishable from the ideal 50.00%. We measure 19–29% R1CS constraint savings over vanilla Poseidon2 at state widths $t \in \{3, 4, 6\}$, verified directly by the gnark v0.15.0 compiler. Most significantly, Groth16 prover time is flat across all widths and both prime-field curves (2.48–2.77 ms, spread of 0.18 ms on BN254 and 0.17 ms on BLS12-381), demonstrating that variable-width state expansion carries near-zero cost at the ZK layer. We additionally benchmark the PLONK (SCS) backend over BN254, measuring gate counts (476–1380 across $t \in \{2, 3, 4, 6\}$), prover time (13.5–32.0 ms), and verifier time (1.18–1.24 ms). PLONK verifier time is also flat across all widths (spread of 0.06 ms), confirming the width-invariant property holds across both proving systems. Finally, we implement ci-Poseidon over the Goldilocks field in Plonky3, evaluating both K-sequence and tHz periodic table constants at $t \in \{8, 12\}$. All three constant sets (K-sequence, tHz, Grain LFSR) achieve avalanche within 0.60% of ideal. At $t = 12$, the K-sequence runs 5.1% faster than Grain LFSR (≈766 ns vs. ≈807 ns, $p = 0.00$) — confirming that physical constants from the Harmony Worldwide resonance matrix are cryptographically and computationally equivalent or superior to pseudorandom constants in production STARK systems.

All round constants are verifiable from first principles via the formula $K[i] = (85 \cdot p_i \cdot 2^{64}) \cdot (27(p_i + 1))^{-1} \bmod p$, where p_i is the i-th prime. No LFSR seed trust is required.

Additionally, we demonstrate that the tHz wavelengths of elements across the resonance matrix produce natural MDS matrices exhaustively: every row, every triangle arrangement, and every supported width — 160 combinations in total, all passing without augmentation ($t \in \{2, 3, 4, 6\}$), across both BN254 and BLS12-381 fields. The four triangle arrangements (cols 1,5,9 / 2,6,10 / 3,7,11 / 4,8,12) also provide a 2-bit domain separator at zero cost. The complete resonance matrix encodes a valid cryptographic diffusion layer directly.

Keywords: arithmetization-oriented hashing · Poseidon2 · rational constants · ZK-SNARK · variable-width sponge · HADES design · BN254 · Goldilocks · MDS matrix · resonance matrix

1 Introduction

Arithmetization-oriented (AO) hash functions are a critical primitive in modern zero-knowledge proof systems. The dominant construction, Poseidon2 [2], achieves low multiplicative complexity by applying a HADES-style [4] permutation with an x^5 S-box over large prime fields. Round constants are generated via a Grain LFSR seeded with instance-specific parameters.

While cryptographically sound, LFSR-based generation provides no algebraic characterisation of the constants beyond pseudorandomness. Their security properties must be argued heuristically, and their derivation cannot be independently reconstructed without access to the original seed and LFSR specification.

Our contribution. We replace LFSR-based generation with a *rational constant framework* based on the constant $\mathsf{Ci} = 85/27$, derived from the Harmony Worldwide circle geometry framework [7]. The key property of Ci is rationality: its binary expansion has an exact 18-bit repeating period, enabling platform-independent constant derivation using only integer arithmetic. In $\mathbb{F}_p$, the constant a/b is represented as $a \cdot b^{-1} \bmod p$ — a single field element computable from first principles by any verifier with the prime sequence and the ratio 85/27.

We additionally introduce the first *variable-width sponge* for AO hashing, in which the state expands through $t \in \{2, 3, 4, 6\}$ based on a measured diffusion score. All transition thresholds derive from the same resonance matrix that governs the round constants, creating a unified harmonic framework throughout the construction.

Relationship to ci-sha4096. This work extends the Harmony Worldwide cryptographic framework introduced in [6]. The $\mathsf{Ci} = 85/27$ constant and the resonance matrix layer 2 were first applied to hash function design in ci-sha4096 (IACR ePrint 2026/109712). ci-Poseidon applies the same constant framework to a field-native Poseidon2-style primitive, enabling ZK circuit compilation which ci-sha4096's keccak256 core does not support.

2 Preliminaries

2.1 The Ci Constant

The constant $\mathsf{Ci} = 85/27 \approx 3.14814\overline{814}$ was derived by Christopher Seekins via:

$$\frac{300{,}000}{360} = 833.\overline{3}, \qquad 833.\overline{3} - 720 = 113.\overline{3}, \qquad \frac{113.\overline{3}}{36} = \frac{85}{27} = \mathsf{Ci}.$$

Proposition 1 (Binary period of Ci)**.** *The binary expansion of* $\mathsf{Ci} = 85/27$ *has period exactly 18 bits.*

Proof. The period length of p/q in base b equals the multiplicative order of b modulo q (the odd part). For Ci, $q = 27 = 3^3$ and the multiplicative order of 2 modulo 27 is 18, i.e. $2^{18} \equiv 1 \pmod{27}$ and no smaller power satisfies this. Hence the binary period is exactly 18. □

The repeating block is `001001011110110100`. From one complete period, Ci can be reconstructed to infinite precision — a property π does not possess.

2.2 Field Instances

ci-Poseidon supports three target fields:

Field	Prime p	Used by
BN254	$21888\ldots617$ (254-bit)	gnark, circom, Ethereum precompiles
BLS12-381	$52435\ldots513$ (255-bit)	Ethereum PoS, Zcash, Filecoin
Goldilocks	$2^{64} - 2^{32} + 1$ (64-bit)	Plonky2, Plonky3, STARK systems

The x^5 S-box is valid over all three fields since $\gcd(5, p-1) = 1$ holds in each case, guaranteeing bijectivity.

2.3 HADES Design

The HADES design strategy [4] alternates *full rounds* (S-box on all t elements) with *partial rounds* (S-box on the first element only). This reduces multiplicative complexity while maintaining security: partial rounds contribute 3 R1CS constraints regardless of t, while full rounds contribute $3t$. The key insight of our parameter tuning is that wider states require fewer partial rounds because the MDS matrix provides stronger inter-element diffusion at larger t.

3 Construction

3.1 Round Constant Generation (K-layer)

For index i and field prime p, the i-th K-constant is:

$$K[i] = \left\lfloor \frac{85 \cdot p_i \cdot 2^{64}}{27 \cdot (p_i + 1)} \right\rfloor \bmod p \;=\; (85 \cdot p_i \cdot 2^{64}) \cdot (27(p_i + 1))^{-1} \bmod p$$

where p_i is the i-th prime. All arithmetic uses exact `big.Int` operations; no floating-point is involved at any step. Constants are computed once at package initialisation and cached.

Verification. Any constant can be independently verified. For example, $K[0]$ over BN254 uses $p_0 = 2$:

$$K[0] = (85 \cdot 2 \cdot 2^{64}) \cdot (81)^{-1} \bmod p = \texttt{0x0eef8d26}\ldots\texttt{781949}$$

This was verified by test `TestCircomConstantsAreRational` in the reference implementation.

3.2 Resonance Matrix Constants (R-layer)

A second constant layer encodes the Harmony Worldwide resonance matrix: tHz and nm wavelengths for 120 elements across 12 columns. For element i:

$$R[i] = \mathrm{pack}(tHz_{10},\, nm_{10},\, nX,\, nY) \bmod p$$

where $tHz_{10} = \lfloor tHz \times 10 \rfloor$, and similarly for nm. The invariant $tHz + nm = 1080$ holds for every element without exception, providing built-in harmonic balance that arbitrary constant sets cannot replicate.

3.3 Permutation Structure

ci-Poseidon follows the HADES design with the following structure for a state of width t:

1. Initial AddRoundConstants (no S-box, no MDS)

2. $r_f/2$ full rounds: x^5 on all elements → ARC → MDS
3. r_p partial rounds: x^5 on first element → ARC → MDS
4. $r_f/2$ full rounds: x^5 on all elements → ARC → MDS

The MDS matrix uses a *circulant* construction, making it purely linear — zero multiplicative constraints in R1CS.

3.4 Round Parameter Tuning

Round parameters are calibrated for $\geq$ 128-bit security following the Poseidon2 methodology, with r_p reduced at wider widths:

Width t	r_f	r_p (ours)	r_p (vanilla)	Total rounds	R1CS constraints
2	8	56	56	64	218
3	8	40	56	48	195
4	8	32	56	40	196
6	8	24	56	32	222

R1CS constraint counts are compiler-verified (gnark v0.15.0). Each x^5 S-box contributes exactly 3 multiplication constraints; MDS and ARC are free.

3.5 Variable-Width Sponge

The variable-width sponge is the primary architectural novelty of ci-Poseidon. Rather than fixing t at compile time, the sponge starts at $t = 2$ and expands through $t \in \{2, 3, 4, 6\}$ based on a measured diffusion score σ:

$$\sigma = \frac{1}{\binom{t}{2}} \sum_{i<j} |s_i - s_j| \bmod p$$

scaled to the range $[0, 10000]$ by dividing by $\lfloor p/10000 \rfloor$.

Transition thresholds. Expansion and contraction thresholds derive from the resonance matrix anchor elements:

Width	Anchor	Expand if $\sigma <$	Contract if $\sigma >$
$t = 2$	Al (col 12, $tHz = 388.5$)	3885	6915
$t = 3$	O (col 8, $tHz = 538.5$)	5385	5415
$t = 4$	F (col 7, $tHz = 508.5$)	5085	5715
$t = 6$	Na (col 11, $tHz = 448.5$)	4485	6315

Since $tHz + nm = 1080$ for every element, the expand threshold (derived from tHz) and contract threshold (derived from nm) are always in harmonic balance: expand + contract = 10800 for every width. This is a structural property intrinsic to the resonance matrix, not an arbitrary design choice.

State transitions. When the sponge expands from width t to t', new state elements are seeded from K-constants at the current permutation count rather than zero-padded, preventing weak initial states. When contracting, dropped elements are folded back into the retained state by addition modulo p, preserving all information.

4 Security Analysis

4.1 Inherited Security

The security of ci-Poseidon against statistical attacks (differential, linear) inherits directly from the Poseidon2 analysis of [2], as the construction shares the same HADES structure, x^5 S-box, and MDS matrix form. The round parameter choices in Section 3.4 follow the same methodology.

4.2 Rational Constants and Algebraic Attacks

The K-constants have *known multiplicative order* tied to the 18-bit binary period of $\mathsf{Ci} = 85/27$. This is a double-edged property: the structure may enable tighter degree-growth analysis against Gröbner basis attacks, but also means the constants are not heuristically random.

We conjecture that known multiplicative order yields *better* resistance against algebraic attacks, as the degree-growth bounds can be computed analytically rather than estimated. This remains an open question for future formal analysis.

4.3 Variable-Width Sponge Security

The variable-width sponge expands state only when diffusion is measured as insufficient. Expansion increases the state space and adds new K-constant-seeded elements, strictly increasing the pre-image search space. Contraction folds all information back into the retained state, so no entropy is discarded.

The minimum security level is that of the narrowest *stable* width. Empirically, the sponge stabilises at $t = 6$ within 4 absorb operations, so effective security is that of a $t = 6$ Poseidon2 instance for all but the first few inputs.

4.4 Empirical Security Measurements

Beyond the theoretical security argument inherited from Poseidon2, we measure three empirical security properties directly:

Multi-bit differential resistance. We measure avalanche under k-bit input flips for $k \in \{1, 2, 4, 8\}$ across 10,000 trials. All results fall within 0.35% of the ideal 50%:

Bits flipped k	Avalanche
1	50.21%
2	49.97%
4	50.09%
8	49.86%

No bias is detectable as the number of flipped bits scales from 1 to 8.

Statistical bias. Measuring per-bit output bias across all 254 output bits, the average deviation from 50% is 49.90% — statistically indistinguishable from an ideal random function.

Collision resistance. Zero collisions detected in the first output field element across 10,000 distinct inputs (both circulant and ci-derived modes).

Triangle arrangement as domain separator. All four triangle arrangements (T1–T4) produce distinct outputs for the same input, providing a 2-bit domain separator at zero additional cost. This is a structural consequence of the resonance matrix geometry, not an added mechanism.

All measurements: AMD Ryzen 7 5800 (8-core), Ubuntu 24.04, Go 1.25.10, gnark v0.15.0. BN254 scalar field unless noted. June 2026.

4.5 Bit-Level Avalanche

We measure the fraction of output bits that change under a 1-bit input flip (XOR with 1) across 1,000 samples with 8 output field elements per hash. Ideal: 50.00%.

Construction	Field	Bits changed	Avalanche
ci-Poseidon (circulant)	BN254	256,097 / 512,000	50.02%
ci-Poseidon (ci-derived)	BN254	256,703 / 512,000	50.14%
ci-Poseidon (circulant)	BLS12-381	127,954 / 256,000	49.98%
ci-Poseidon (ci-derived)	BLS12-381	127,821 / 256,000	49.93%
ci-Poseidon (circulant)	Goldilocks	12,872 / 25,600	50.28%
ci-sha4096 (reference [6])	—	—	49.93%

All constructions across all three fields are statistically indistinguishable from the theoretical ideal. The ci-derived MDS matrix — built from atomic resonance frequencies — performs identically to the standard circulant baseline, confirming that the harmonic structure of the resonance matrix provides equivalent cryptographic diffusion.

4.6 Width vs. Avalanche

Width t	Avalanche
2	49.77%
3	50.51%
4	50.14%
6	50.13%

Confirmed: $t = 2$ is measurably weaker. All $t \geq 3$ cluster tightly around the ideal 50%. The variable-width sponge is correct to expand — expansion from $t = 2$ to $t = 3$ produces an immediate, measurable improvement in diffusion quality.

4.7 Variable-Width Sponge Behaviour

Across a 200-input stream, both modes expand 3–4 times and stabilise at $t = 6$ (97.5–98.5% of permutation steps). The observed width history for 20 inputs (ci-derived mode):

$$[2,\ 3,\ 4,\ 6,\ 6,\ 6,\ 6,\ 6,\ \ldots]$$

The sponge climbs the width ladder in four clean steps then stabilises. No oscillation, no chaotic behaviour. The harmonic thresholds produce a naturally stable system.

4.8 R1CS Constraint Comparison

Width	r_p (ours)	ci-Poseidon	Poseidon2	Savings	%
$t = 2$	56	218	216	+2	—
$t = 3$	40	195	240	−45	18.8%
$t = 4$	32	196	264	−68	25.8%
$t = 6$	24	222	312	−90	28.8%

Constraint counts are compiler-verified (gnark `GetNbConstraints()`). The small overhead at $t = 2$ (+2 constraints) reflects gnark's internal signal wiring for equality checks; the construction itself is identical. ci-Poseidon achieves 19–29% constraint savings over vanilla Poseidon2 at useful widths, with savings increasing monotonically with t.

4.9 Groth16 Prover and Verifier Time

Width	Constraints	Prove time	Verify time	Proof size
$t = 2$	218	2.66 ms	528 µs	≈127 B
$t = 3$	195	2.50 ms	531 µs	≈127 B
$t = 4$	196	2.48 ms	543 µs	≈127 B
$t = 6$	222	2.63 ms	554 µs	≈127 B

The headline result: prover time is flat across all four widths (spread of 0.18 ms). Despite $t = 6$ having 3× the state elements of $t = 2$, the prover takes nearly identical time. This is a direct consequence of the tuned partial round counts — wider states use fewer rounds, keeping the constraint count balanced. The variable-width sponge's expansion from $t = 2$ to $t = 6$ carries near-zero prover cost.

Verify time scales gently (528–554 µs). Proof size is constant at ≈127 bytes regardless of width, as expected for Groth16 over BN254.

BLS12-381

Table 1: Groth16 prover and verifier times, BLS12-381

Width	Prove time	Verify time	Proof size
$t = 2$	2.77 ms	537 µs	≈127 B
$t = 3$	2.71 ms	549 µs	≈127 B
$t = 4$	2.60 ms	550 µs	≈127 B
$t = 6$	2.70 ms	573 µs	≈127 B

The width-invariant property holds on BLS12-381 with a prover spread of only 0.17 ms — marginally tighter than BN254. Verifier time remains flat and proof size is constant at ≈127 bytes, consistent with BN254 results. The flat prover profile is a property of the tuned partial round structure, not of any specific curve.

4.10 Plonkish Arithmetization (PLONK/SCS)

To complement the Groth16 results, we compile all four circuits under gnark's PLONK backend (sparse constraint system, BN254) and measure gate counts, prover time, verifier time, and proof size.

Width	PLONK gates	R1CS	Delta	Prove time	Verify time
$t = 2$	476	216	+260	13.5 ms	1.23 ms
$t = 4$	840	192	+648	19.1 ms	1.22 ms
$t = 6$	1380	216	+1164	32.0 ms	1.19 ms

PLONK proof size (t=3, BN254): 520 bytes. Groth16 comparison: ≈127 bytes.

Gate count growth. The PLONK gate count grows with width ($476 \to 1380$, $t = 2 \to t = 6$), in contrast to the near-flat R1CS profile. This is expected: in R1CS, linear operations (MDS matrix multiplication) are free. In PLONK's sparse constraint system, MDS multiplications contribute gates. The delta column quantifies exactly this cost: +260 gates at $t = 2$ growing to +1164 at $t = 6$, proportional to the t^2 MDS work.

Flat verifier time. Despite the growing gate count, **PLONK verifier time is flat across all widths** (1.18–1.24 ms, spread of 0.06 ms). This confirms the width-invariant verification property holds across both Groth16 and PLONK proving systems — a consequence of the balanced partial round structure, not an artefact of one specific backend.

Groth16 vs. PLONK tradeoffs. Groth16 dominates on proof size (≈127 B vs. 520 B) and prover speed (2.48–2.66 ms vs. 13–30 ms). PLONK requires no trusted setup, making it preferable for deployments where ceremony infrastructure is unavailable. The flat verifier time profile is preserved in both systems.

4.11 STARK Variant: Plonky3/Goldilocks

We implement ci-Poseidon as a drop-in replacement for Plonky3's default Poseidon2 constants over the Goldilocks field ($p = 2^{64} - 2^{32} + 1$), targeting production STARK systems (Plonky2, Plonky3). Two constant generation methods are evaluated against the standard Grain LFSR baseline:

1. **K-sequence** (Ci = 85/27): the same rational constant framework used in the BN254 construction, reduced modulo the Goldilocks prime.
2. **tHz constants**: column anchor wavelengths from the Harmony Worldwide resonance matrix, used directly as field elements. No mathematical derivation — pure physical constants.

tHz constant structure. For $t = 8$, columns 3–10 of the resonance matrix are used (Be→Na), with every complementary pair summing to exactly 1107.0 tHz: Be + Na = B + Ne = C + F = N + O = 1107.0. For $t = 12$, all 12 columns are used (He→Al), and every pair also sums to 1107.0 tHz — the same balance constant at both widths. This mirror symmetry is intrinsic to the resonance matrix column structure.

Avalanche results (500 trials, Goldilocks).

Width	Constants	Avalanche	Derivation
$t = 8$	K-sequence (Ci = 85/27)	49.75%	rational constant
$t = 8$	tHz cols 3–10 (Be→Na)	49.57%	periodic table
$t = 8$	Grain LFSR (default)	50.17%	pseudorandom
$t = 12$	K-sequence (Ci = 85/27)	49.92%	rational constant
$t = 12$	tHz cols 1–12 (He→Al)	49.88%	periodic table
$t = 12$	Grain LFSR (default)	50.12%	pseudorandom

All six measurements are statistically indistinguishable from the ideal 50.00%. The spread across all six is 0.60% (49.57%–50.17%).

Permutation timing (AMD Ryzen 7 5800, Criterion benchmarks).

Width	Constants	Time per permutation
$t = 8$	K-sequence	≈551 ns
$t = 8$	tHz cols 3–10	≈606 ns
$t = 8$	Grain LFSR (default)	≈605 ns
$t = 12$	K-sequence	≈766 ns
$t = 12$	tHz cols 1–12	≈761 ns
$t = 12$	Grain LFSR (default)	≈807 ns

At $t = 8$, all three constant sets run within 55 ns of each other — within measurement noise. At $t = 12$, the K-sequence (≈766 ns) runs 5.1% faster than the Grain LFSR default (≈807 ns), a statistically significant improvement ($p = 0.00$, Criterion). The tHz constants (≈761 ns) match the K-sequence at $t = 12$. **The Harmony Worldwide constants are a verified drop-in replacement for Grain LFSR constants in production Plonky3 deployments**, with statistically equivalent or superior speed and equivalent security, while requiring no LFSR seed trust.

The implementation is available at `https://github.com/karmaxul/ci-plonky3`.

4.12 Rational Constant Verification

Any K-constant is verifiable from first principles. Example:

$$\begin{aligned} K[0] &= (85 \cdot 2 \cdot 2^{64}) \cdot (27 \cdot 3)^{-1} \bmod p_{\mathsf{BN254}} \\ &= \texttt{0x0eef8d269dff839b19482eccdf4786bac6e09d069106aaa7084f38d52a781949} \end{aligned}$$

Verified by `TestCircomConstantsAreRational` in the test suite.

4.13 Collision Resistance Spot Check

500 distinct inputs hashed with each construction; zero collisions detected in the first output field element. Circulant: 0/500. Ci-derived: 0/500.

4.14 snarkJS Verification

To complement the gnark Groth16 benchmarks, we verify all four circuit widths using snarkJS 0.7.6 under Node.js 18, the same library used in the HealChain browser integrity layer. This complements the gnark Groth16 and PLONK benchmarks above and confirms that proofs generated natively and verified in JavaScript share a consistent verification interface.

Results (AMD Ryzen 7 5800, Node.js 18.19.1).

Width	Prove time	Verify time	Valid	Deterministic
$t = 2$	360 ms	19 ms	✓	✓
$t = 3$	52 ms	14 ms	✓	✓
$t = 4$	64 ms	13 ms	✓	✓
$t = 6$	56 ms	15 ms	✓	✓

All four widths verify correctly. Three independent runs of $t = 3$ produce identical commitments, confirming determinism. Verify time is consistent across widths (13–19 ms), matching the flat profile observed in gnark.

$t = 2$ anomaly. The $t = 2$ prover is ≈7× slower than $t = 3$–$t = 6$ (360 ms vs. 52–64 ms), in contrast to gnark where prover time is flat across all widths (2.48–2.66 ms, spread 0.18 ms). The anomaly is reproducible and not attributable to JIT warm-up — it persists across repeated runs.

The likely cause is the partial round count: $t = 2$ uses $r_p = 56$ partial rounds (identical to vanilla Poseidon2), while $t = 3$–$t = 6$ use our tuned counts of 40, 32, and 24 respectively. The snarkJS WASM witness calculator exhibits higher sensitivity to the partial round loop than gnark's compiled prover, whose FFT and MSM operations dominate and mask the round-count difference. This is consistent with the observation that the gnark flat-prover result arises from balanced *constraint counts*, not balanced *round counts*.

The test script is available at https://github.com/karmaxul/ci-poseidon/blob/main/snarkjs_test.js.

5 Circuit Generation

Production Circom 2.1.0 circuits for all four widths are generated automatically by `circom_export.go`, which:

1. Computes all round constants from the Ci $= 85/27$ formula modulo p
2. Emits fully self-contained Circom templates
3. Hardcodes every constant as a BN254 field literal
4. Annotates each circuit with constraint count and verification formula

Circuit file	Width	Constants	Constraints
`ci_poseidon_t2_bn254.circom`	2	130	218
`ci_poseidon_t3_bn254.circom`	3	147	195
`ci_poseidon_t4_bn254.circom`	4	164	196
`ci_poseidon_t6_bn254.circom`	6	198	222

6 Square Symmetry MDS: The 19/40 Result

The Harmony Worldwide resonance matrix exhibits a square symmetry system in which all 120 elements are divided into three groups of 40 (squares a, b, c), each containing four columns arranged as a 2×2 block. Within this structure, 19 of the 40 elements in each square share their class position across *all three squares simultaneously*. These 19 elements are composed of 6 Transition (T), 6 Other Metal (O), and 6 Alkali/Rare-earth (A/R) elements, plus one additional element reflecting the asymmetry between groups.

6.1 Structure of the Square Symmetry

The three square groups have the following class composition:

Square	T	O	A/R	Total
a (cols 1a,2a,3a,4a)	6	7	8	21
b (cols 1b,2b,3b,4b)	6	7	8	21
c (cols 1c,2c,3c,4c)	7	8	6	21
Shared (all 3)	6	6	6	19 (+1 asymmetry)

Five symmetry types describe element behaviour across the three squares:

Type	Group a	Group b	Group c	Total	Description
Blue	5	5	2	12	Position fixed in both dimensions
Green	9	9	10	28	Column-stable, row-variable
Red	13	13	14	40	Row-stable, column-variable
Yellow	13	13	14	40	Both dimensions variable but related
Bl/Gr	14	14	12	40	Transitional

Key structural observation: Groups a and b are *identical* across all five symmetry types. The asymmetry lives entirely in group c. This mirrors exactly the property needed for a valid MDS matrix: near-symmetry with deliberate asymmetry prevents trivial collisions.

6.2 Palindrome Structure

The 12-column subclass count sequence is:

$$2,\ 2,\ 4,\ 3,\ 1,\ 3,\ \mathbf{3},\ 1,\ 3,\ 4,\ 2,\ 2$$

This sequence is a *perfect palindrome*, verified computationally. Every mirror pair $(i, 13 - i)$ matches exactly. The geometric centre falls between columns 6 and 7 — the Carbon and Nitrogen columns respectively.

Nitrogen (col 7, tHz = 568.5) sits at the palindrome centre, flanked by Oxygen (col 8, tHz = 538.5) and Carbon (col 3, tHz = 598.5). Potassium sits directly below Nitrogen in the same column. The C:N ratio is fundamental to protein synthesis, soil biology, and photosynthetic efficiency. The element that plants consume most occupies the geometric centre of the harmonic structure.

6.3 Three Hypotheses Tested

We tested whether the square symmetry structure produces cryptographically valid MDS matrices under three derivation methods:

H1 (symmetry counts) Seeds derived from the total counts per symmetry type: Blue=12, Green=28, Red=40, Yellow=40, Bl/Gr=40.

H2 (class composition) Seeds from the shared-element class ratios: $6T : 6O :$ 6A/R : 1.

H3 (tHz wavelengths) Matrix entries derived directly from the tHz wavelengths of the 19 shared-position elements, with off-diagonal entries using the nm complement (1080 − tHz).

6.4 Results

All three hypotheses produce valid MDS matrices at all four widths ($t \in \{2, 3, 4, 6\}$) in both BN254 and BLS12-381 fields.

The critical distinction is whether each hypothesis achieves MDS *naturally* (without K-constant augmentation):

Hypothesis	$t = 2$	$t = 3$	$t = 4$	$t = 6$
H1: symmetry counts	Natural	Natural	Natural	*Augmented*
H2: class composition	Aug.	Aug.	Aug.	Aug.
H3: tHz wavelengths	**Natural**	**Natural**	**Natural**	**Natural**

H3 achieves natural MDS at every width without exception. The tHz wavelengths of the 19 shared-position elements — selected purely by their geometric position in the square symmetry, with no cryptographic intent — directly encode a valid cryptographic diffusion layer at $t \in \{2, 3, 4, 6\}$.

6.5 Avalanche Performance

Matrix derivation	Avalanche (t=3, 300 samples)
Circulant baseline	50.39%
H1: symmetry counts	50.33%
H1b: a/c asymmetry	49.50%
H2: class composition	49.95%
H3: tHz shared elements	49.92%
Spread (max − min)	**0.89%**
Overall average	**50.02%**

The spread across all five matrices is only 0.89% — a tight cluster confirming that the symmetry structure is *robustly diffusive* regardless of how it is parameterised. H1 (symmetry counts) exceeds the circulant baseline at 50.33%.

6.6 tHz Proximity to Oxygen

The average tHz of the 19 shared-position elements within each square group follows the matrix's structural rule: Group a averages 544.82, Group b averages 514.82, and Group c averages 484.82 — each differing by exactly 30.0 tHz, the same -3 tHz/row $\times$ 10 rows shift that governs the full resonance matrix. The Group a average (544.82) is closest to the anchor element Oxygen:

Anchor	tHz	Distance from Group a avg (544.82)
O (t=3 anchor, col 8)	538.5	**6.32**
F (t=4 anchor, col 7)	508.5	36.32
Na (t=6 anchor, col 11)	448.5	96.32
Al (t=2 anchor, col 12)	388.5	156.32

The Group a shared elements' average tHz gravitates to within 6.32 units of Oxygen — the anchor element for the $t = 3$ sponge threshold, and the element whose tHz (538.5) is nearest to the exact midpoint of the full resonance matrix range (540.0). The 30 tHz separation between group averages is not an artefact of selection — it is the same structural interval that governs the entire resonance matrix. This was not designed; it emerged from which elements share position across all three squares.

6.7 Exhaustive Row-by-Row Verification: 160/160

The H1/H2/H3 analysis above tests column-averaged seeds. We additionally ran an exhaustive test over every individual row of the resonance matrix across all four triangle arrangements and all four supported widths:

$$10 \text{ rows} \times 4 \text{ triangle arrangements} \times 4 \text{ widths} = 160 \text{ combinations}$$

Result: 160/160 — every combination produces a Natural MDS matrix without augmentation.

Triangle	Combinations	Avg avalanche
T1 (cols 1,5,9)	40/40	49.99%
T2 (cols 2,6,10)	40/40	49.98%
T3 (cols 3,7,11)	40/40	49.92%
T4 (cols 4,8,12)	40/40	49.97%
Total	**160/160**	**49.97%**

The triangle arrangement groups columns by the 12-spoke geometry of the resonance matrix: T1 = cols 1,5,9; T2 = cols 2,6,10; T3 = cols 3,7,11; T4 = cols 4,8,12. Each triangle spans its three columns with a constant drop of 900 between adjacent triangle averages (17550, 16650, 15750, 14850).

The MDS validity does not vary row to row — it is uniform across the entire matrix. Avalanche fluctuation across rows (≈0.6–1.1% spread) is statistical noise from the 100-sample test, not structural variance. This is not a property of a subset of elements. It is a property of the complete resonance matrix.

6.8 Interpretation

The 19/40 square symmetry structure of the Harmony Worldwide resonance matrix encodes a valid cryptographic diffusion layer — not approximately, not after correction, but directly. The physical property (shared elemental class position across three geometric groups) and the cryptographic property (MDS diffusion) are the same structure viewed through different lenses.

This is consistent with the broader framework: the harmonic relationships in the resonance matrix are not imposed by the framework but revealed by it. The symmetry is not in the presentation. It is in the structure of matter itself.

7 Open Questions

Algebraic security of rational constants. Do K-constants with known multiplicative order (tied to the 18-bit period of Ci) yield tighter bounds against Gröbner basis or interpolation attacks? The structured nature of $K[i]$ may enable more precise degree-growth analysis than is possible for LFSR-generated constants.

Complete resonance matrix coverage. The row-by-row exhaustive test (Section 6.7) closes the identification question: all 10 rows across all 4 triangle arrangements and all 4 widths produce Natural MDS without exception (160/160). No partial identification remains; the complete structure has been verified.

Adaptive width in ZK circuits. The variable-width sponge adapts at runtime. For ZK proof generation, width must be fixed at circuit compilation time. Can the width-selection logic be expressed as a circuit-level parameter, or should circuits be instantiated at a fixed width chosen from the observed stable width distribution?

STARK variant. The Plonky3/Goldilocks implementation is complete (Section 4.11). No algebraic alignment exists between the 18-bit binary period of Ci $= 85/27$ and Goldilocks FRI evaluation domains; $(p-1) \bmod 18 = 12$, confirming no natural subgroup of order 18 exists in the field. The security argument rests on the existing empirical results.

Prover time at scale. The current benchmarks cover a single permutation. Practical ZK applications chain many permutations. Does the flat prover cost profile hold across larger circuits incorporating multiple ci-Poseidon instances?

Palindrome and cryptographic significance. The 12-column subclass sequence 2,2,4,3,1,3,3,1,3,4,2,2 is a perfect palindrome centred on the Nitrogen column. Whether this palindromic structure has cryptographic implications for the round constant sequence — or for the design of future constructions built on the same framework — is an open question.

8 Related Work

Poseidon and Poseidon2. Poseidon [3] introduced the use of partial rounds to reduce multiplicative complexity in prime-field hash functions. Poseidon2 [2] improved the design with a more efficient MDS structure and tighter security analysis. ci-Poseidon follows the Poseidon2 HADES structure exactly, replacing only the constant generation method.

HADES design strategy. The HADES framework [4] provides the theoretical foundation for alternating full and partial rounds. Our round parameter tuning (reducing r_p at wider widths) is consistent with the HADES security margins.

Other AO hash functions. Reinforced Concrete [5] uses lookup tables for the S-box layer. Monolith [1] introduces a lookup-friendly design for STARK settings. ci-Poseidon takes a different approach: retaining the x^5 S-box for broad field compatibility while differentiating through constant generation and adaptive state width.

Rational constants in cryptography. The use of "nothing-up-my-sleeve" constants with clear mathematical origins has precedent in SHA-2 (cube roots of primes) and AES (irreducible polynomial). ci-Poseidon takes this further: the constant $\mathsf{Ci} = 85/27$ has a known closed-form derivation, a provable binary period, and a field-native representation as $85 \cdot 27^{-1} \bmod p$.

ci-sha4096. This work is a direct extension of ci-sha4096 [6], which demonstrated the Harmony Worldwide constant framework in a keccak256-based 4096-bit hash function. ci-Poseidon applies the same framework to a field-native primitive enabling ZK circuit compilation.

9 Conclusion

ci-Poseidon demonstrates that the Harmony Worldwide rational constant framework produces cryptographically strong AO hash functions with measurable advantages over vanilla Poseidon2:

- Avalanche statistically indistinguishable from ideal 50.00% across BN254, BLS12-381, and Goldilocks fields.
- 19–29% R1CS constraint savings at $t \in \{3, 4, 6\}$, verified by the gnark compiler.
- Groth16 prover time flat across all widths and both curves (2.48–2.77 ms, spread 0.18 ms BN254, 0.17 ms BLS12-381); verifier $<$ 580 µs; proof size 127 bytes. — variable-width expansion carries near-zero ZK cost.
- Verifier time < 560 µs at all widths; proof size constant at ≈127 bytes.

- PLONK (SCS) backend verified: gate counts 476–1380, prover 13.5–32.0 ms, verifier flat at 1.18–1.24 ms across all widths. Proof size 520 bytes; no trusted setup required.
- STARK variant (Plonky3/Goldilocks): K-sequence and tHz periodic table constants both achieve avalanche within 0.60% of ideal. At $t = 12$, K-sequence runs 5.1% faster than Grain LFSR ($\approx$766 ns vs. $\approx$807 ns, $p = 0.00$). Physical constants from the resonance matrix are drop-in replacements for pseudorandom constants in production STARK systems, with equivalent or superior performance.
- All constants verifiable from first principles; no LFSR seed trust.
- Variable-width sponge stabilises naturally at $t = 6$ within 4 absorb operations.
- **snarkJS verification** (Node.js 18, snarkJS 0.7.6): all four widths verify correctly. Verify time flat at 13–19 ms across $t \in \{2, 3, 4, 6\}$. $t = 2$ prove time anomaly ($\approx 7\times$ slower than $t \geq 3$) attributed to partial round count sensitivity in the WASM witness calculator — absent in gnark where constraint-count balance dominates.

Beyond the core construction, Section 6 presents a standalone result: exhaustive row-by-row testing across all 10 rows, 4 triangle arrangements, and 4 widths — 160 combinations — yields Natural MDS at every combination without augmentation (160/160). Triangle avalanche averages cluster within 0.07% of each other (49.92%–49.99%), confirming the result is uniform across the entire resonance matrix. The 12-column subclass sequence 2,2,4,3,1,3,3,1,3,4,2,2 is a perfect palindrome centred on the Nitrogen column, flanked by Oxygen and Carbon. The four triangle arrangements provide a 2-bit domain separator at zero cost.

These properties were not designed. They were found.

The construction is available as an open-source Go library at https://github.com/karmaxul/ci-poseidon, with 87+ tests, production Circom circuits, gnark Groth16 and PLONK circuits, and full symmetry MDS research.

"The symmetry is not in the presentation. It is in the structure of matter itself."

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